Assignment: Adaptive Fuzzy Control System with Programming Implementation

Course: Fuzzy Logic and Control Systems

Objective

Design, implement, simulate, and optimize an adaptive fuzzy control system using programming (Python, MATLAB, or C++). The system must respond to real-world conditions and adapt based on environmental and process changes. All design steps, decisions, evaluations, and justifications must be explained clearly.

Assignment Scenario

You are assigned to develop a **Smart Greenhouse Adaptive Fuzzy Climate Control System**. The greenhouse grows multiple plant species, each with different temperature and humidity needs that vary with their growth stages.

Inputs (Minimum 3 Required)

- Temperature (°C)
- Humidity (%)
- Plant Growth Stage (Seedling, Vegetative, Flowering)

Outputs (Minimum 2 Required)

- Heater/Cooling Fan Power Level (0%-100%)
- Misting System Intensity (0%-100%)

Part 1: System Modeling (Manual Design Required)

1. Select at least three different plant species.

2. For each plant and each growth stage, define the ideal temperature and humidity ranges.
3. Create a comparison table showing differences in climate needs.
4. Explain why fuzzy logic is better than crisp logic or PID controllers for this system.

Part 2: Fuzzy System Design

1. Define fuzzy membership functions for all input variables (minimum 5 fuzzy sets per input).
2. Design **two separate controllers**:
   (a) Mamdani-based fuzzy controller
   (b) Takagi-Sugeno-based fuzzy controller
3. Create at least 25 fuzzy rules for each controller.
4. Ensure proper fuzzification and defuzzification methods are selected and justified.

Part 3: Programming Implementation

Implement both Mamdani and Sugeno controllers using one of the following:

- Python (with `skfuzzy` or full custom code)
- MATLAB (using FIS tools or custom implementation)
- C++ (no use of fuzzy libraries; full manual coding required)

Part 4: Dynamic Adaptation (Adaptive Feature)

1. Implement a mechanism to automatically adjust fuzzy sets and rules when plant type or growth stage changes.
2. Clearly explain the adaptation logic.
3. Plot graphs of control outputs over time under different plant and weather conditions.

Part 5: Performance Evaluation (Simulation Required)

Conduct at least 20 random test simulations with changing weather conditions and compare the performance of both controllers using the following table:

Controller Type	Avg Response Time	Avg Error	Energy Usage	Smoothness Score
Mamdani				
Sugeno				

Analyze which controller performs better and provide reasons.

Part 6: Optimization

Use any optimization technique (Genetic Algorithm, PSO, etc.) to improve fuzzy membership parameters. Present results before and after optimization.

Part 7: Final Report (Detailed Justification Required)

Your final report must include:

- System description and problem justification
- Membership function design reasoning
- Rule creation explanation
- Code implementation overview
- Adaptation strategy
- Performance comparison (Mamdani vs Sugeno)
- Optimization impact
- Real-world limitations and future improvements

Bonus (Optional for Extra Marks)

- Create a GUI interface for live simulation.
- Add reinforcement learning to evolve fuzzy rules.